SPI PROTOCOL

Features

- The acronym SPI stands for “Serial Peripheral Interface” – is a synchronous serial communication protocol.
- Was suggested by Motorola.
- It is sort of a “De Facto” standard.
- Supports ONLY single Master Multi-Slave communication , in which master is ALWAYS ANY CPU & slaves mostly Peripheral Devices But may also be another CPU.
- Originally supports 4 wired bus, But Since not a “De Facto” Standard may from slave to slave as 2 wired/3 wired also.

- The four wires/lines of SPI Bus are

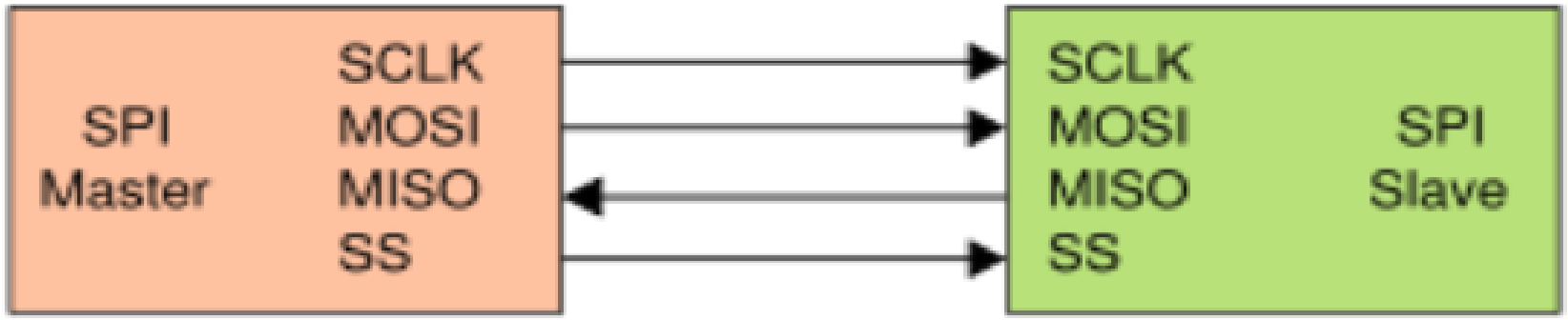
  SS/CS (active low)

  MOSI/(some other acronym)

  MISO/(some other acronym)

  CLK/SCLK/SCK
- Originally Supports FULL Duplex Communication, But Since Not a De Facto Standard may vary as simplex/half Duplex also.
- Support Bus Speeds <= 10MHz/ Mbps or more also
- Originally Not Support for Handshaking/acknowledgement signal/states But Since not a “De Facto” Standard may vary from slave to slave.

- Support Serial Data transfer Order Either MSB first or LSB first or BOTH with respect to slave.
- Supports 4 modes of Serial Communication depending on parameter “clock signal” interpreted in 2 ways namely “clock polarity (CPOL)”-(denotes initial/start/idle state of clock line for a communication session) & “clock phase (CPHA)”-(denotes high to low/low to high transitions) which decide “Data Validity & Transition of serial bit” in SPI communication.



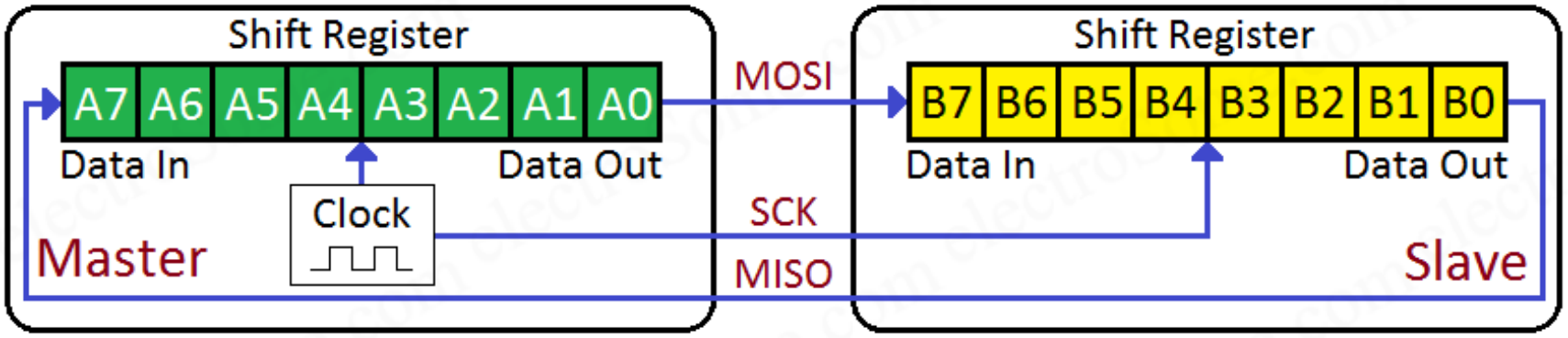
SPI bus: single master and single slave

The SPI bus specifies four logic signals.

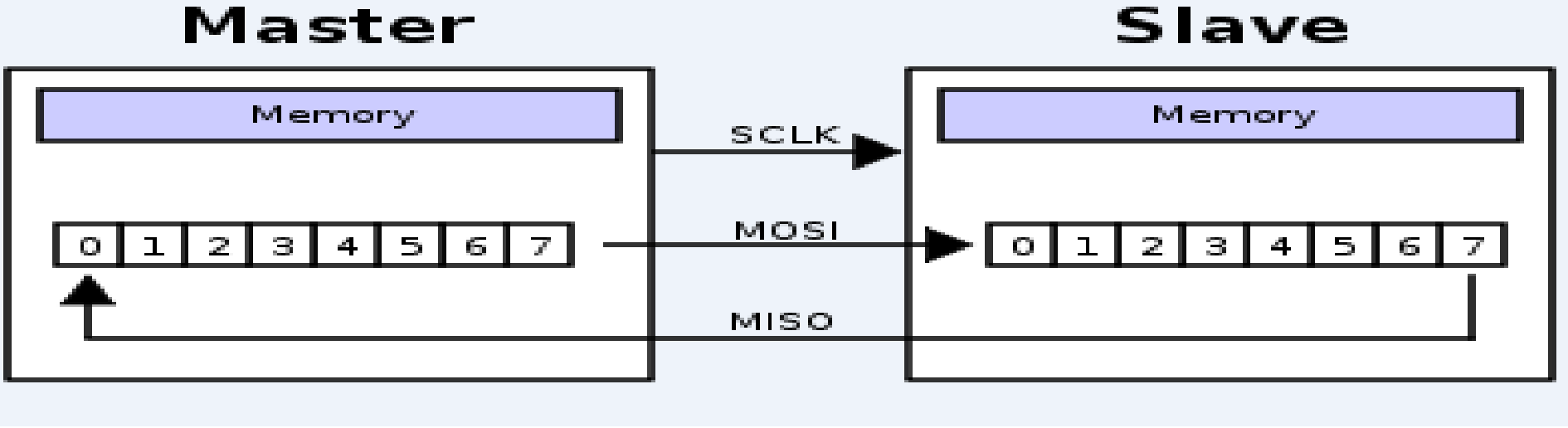
- SCLK — Serial Clock (output from master)
- MOSI — Master Output, Slave Input (output from master)
- MISO — Master Input, Slave Output (output from slave)
- SS — Slave Select (active low; output from master)

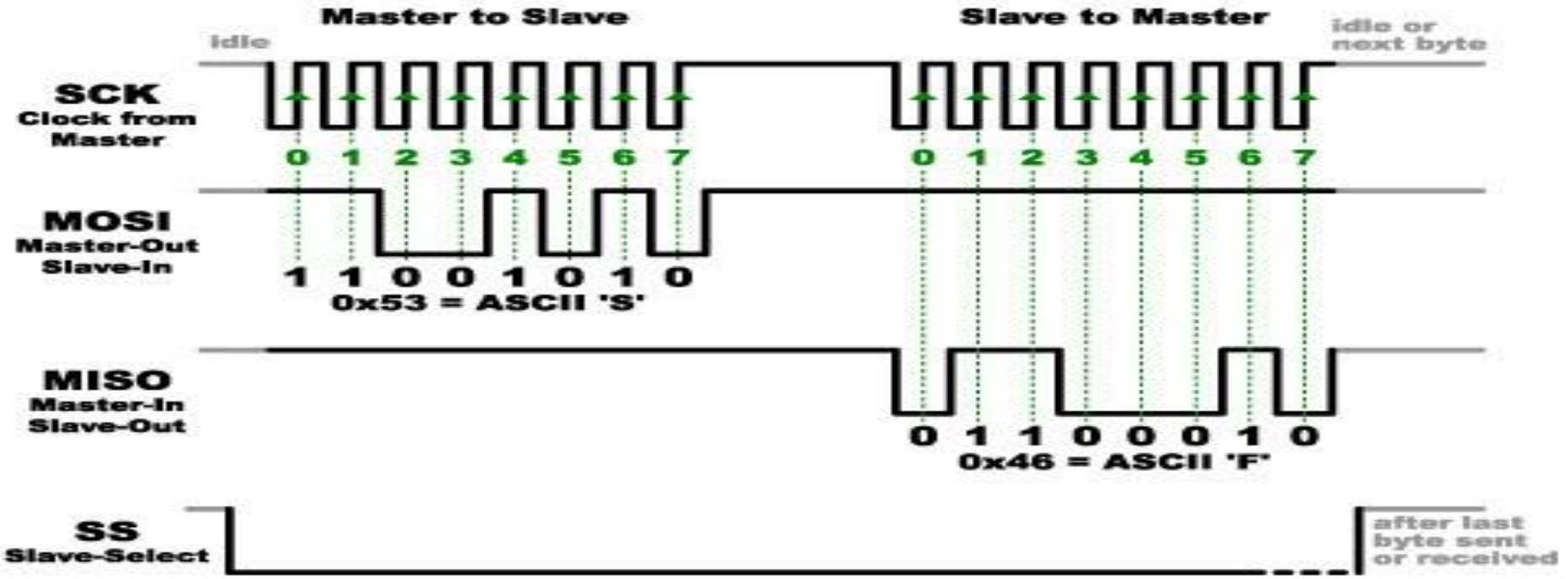
Sometimes, the following naming convention is used:

- SCLK — Serial Clock (output from master)
- SDI — Serial Data In
- SDO — Serial Data Out
- CS — Chip Select (active low; output from master)


Shift Register
A7
A6
A5
A4
A3
A2
A1
A0
Data In
Data Out
Clock
Master
MOSI
SCK
MISO
Shift Register
B7
B6
B5
B4
B3
B2
B1
B0
Data In
Data Out
Slave

How SPI Originally Works

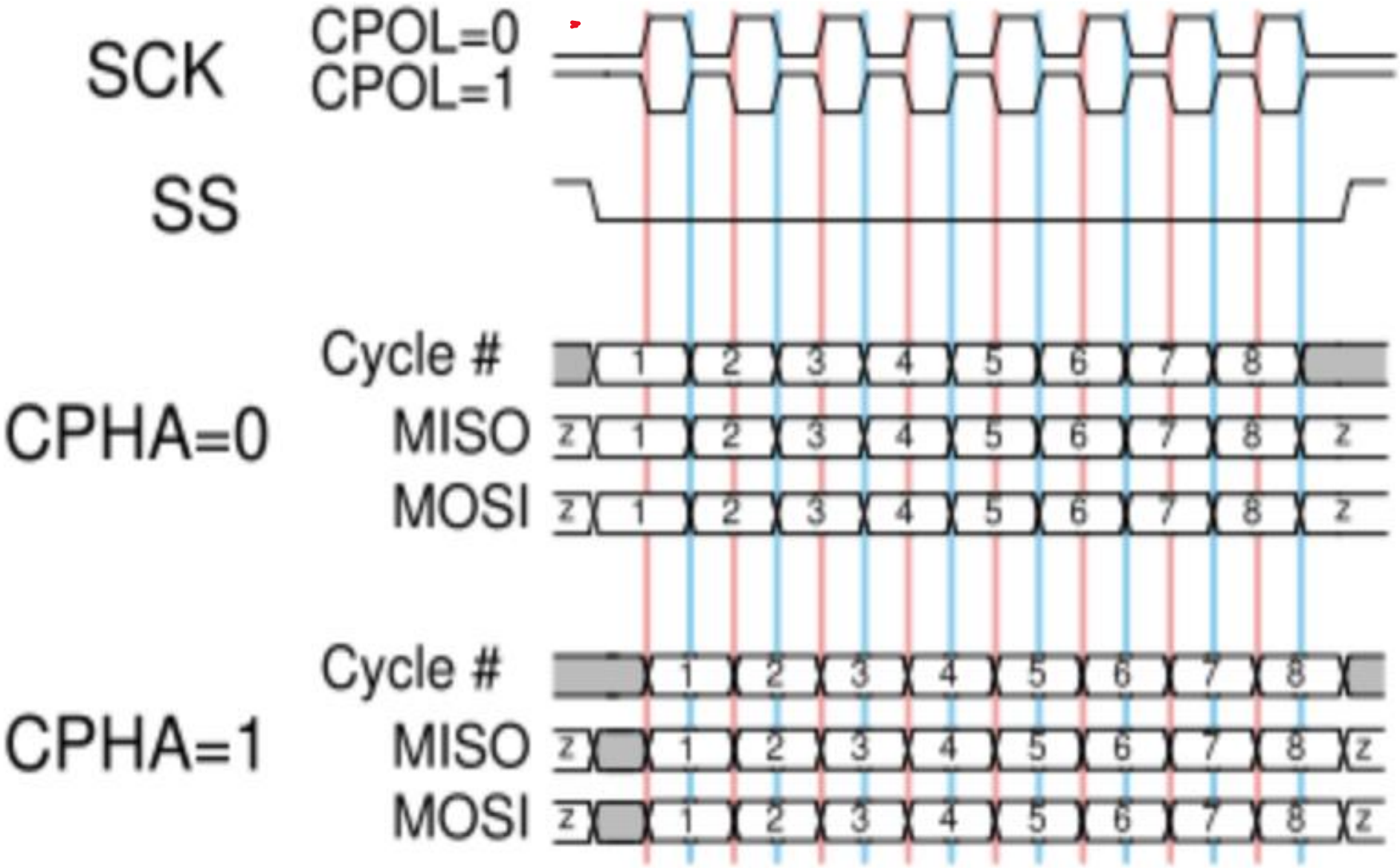




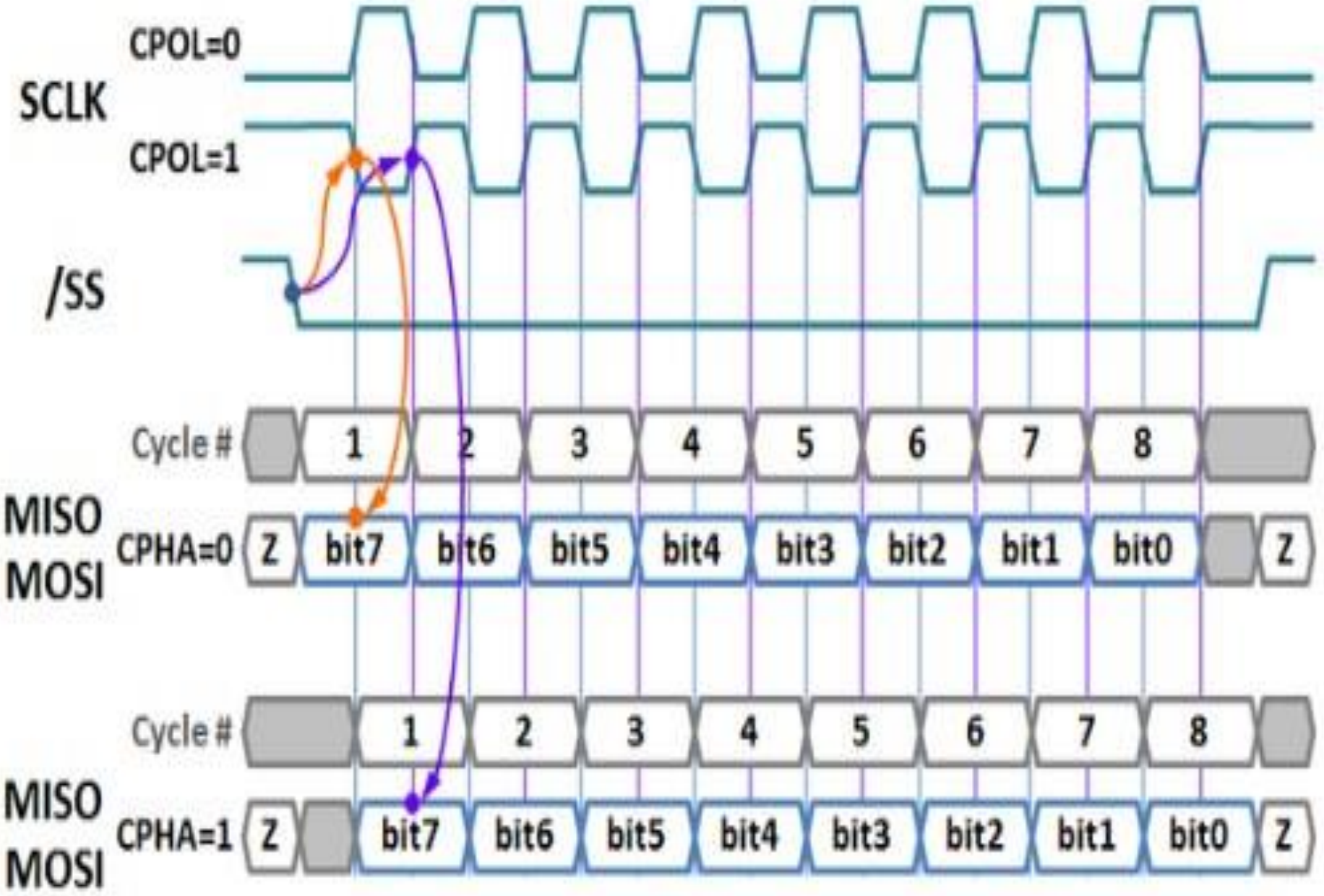
Refer SPI Working PPTS

Animation with Explaination

Clock polarity and phase



SPI Timing Chart



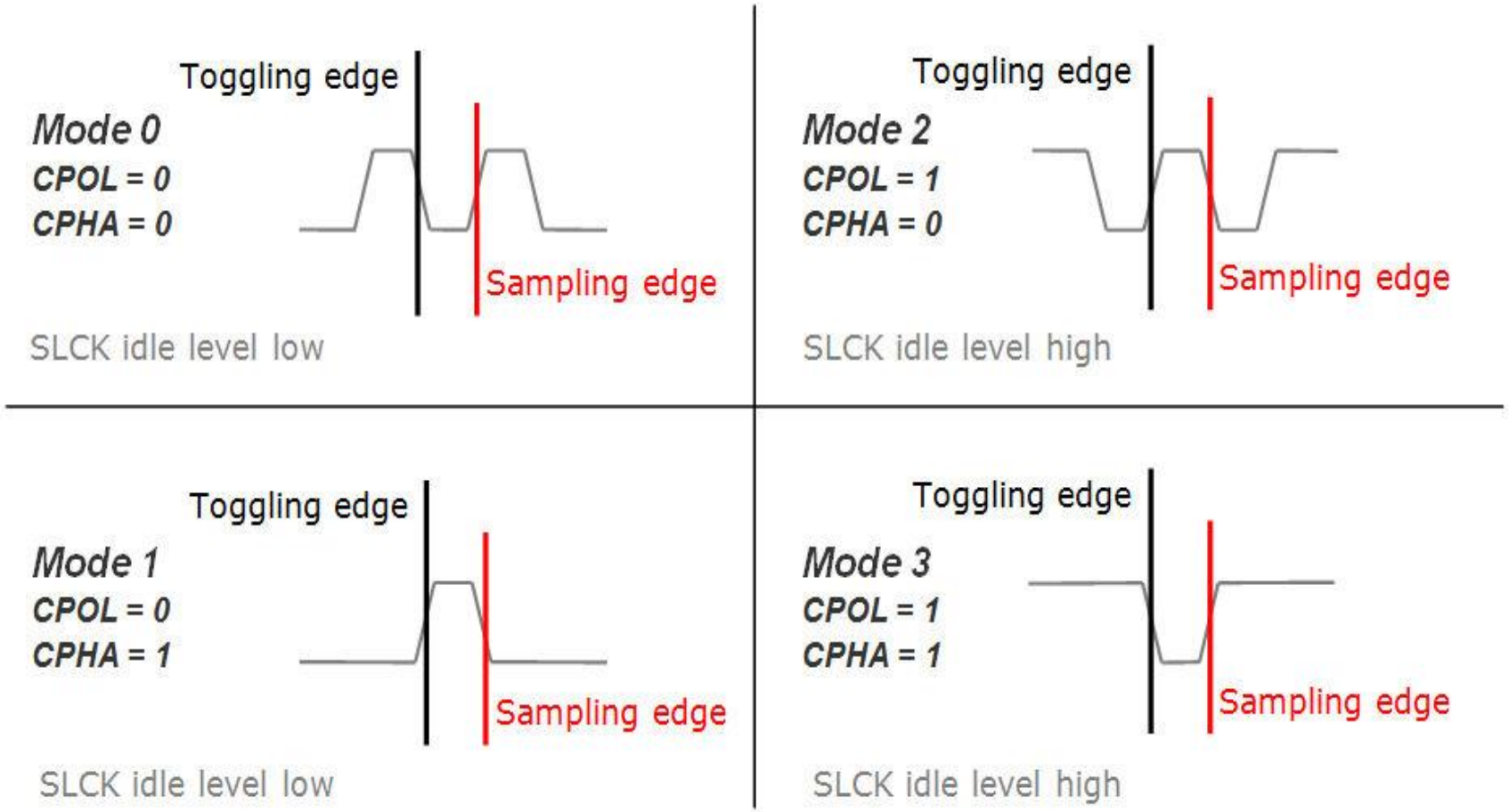


Figure 3 : SPI modes are defined with the parameters 'CPOL' – clock polarity and 'CPHA' – clock phase, which explicitly define 3 parameters: the edges used for data sampling and data toggling and the SCL clock signal idle level – that is the conventional level SCLK is set at when the bus is not in communication.

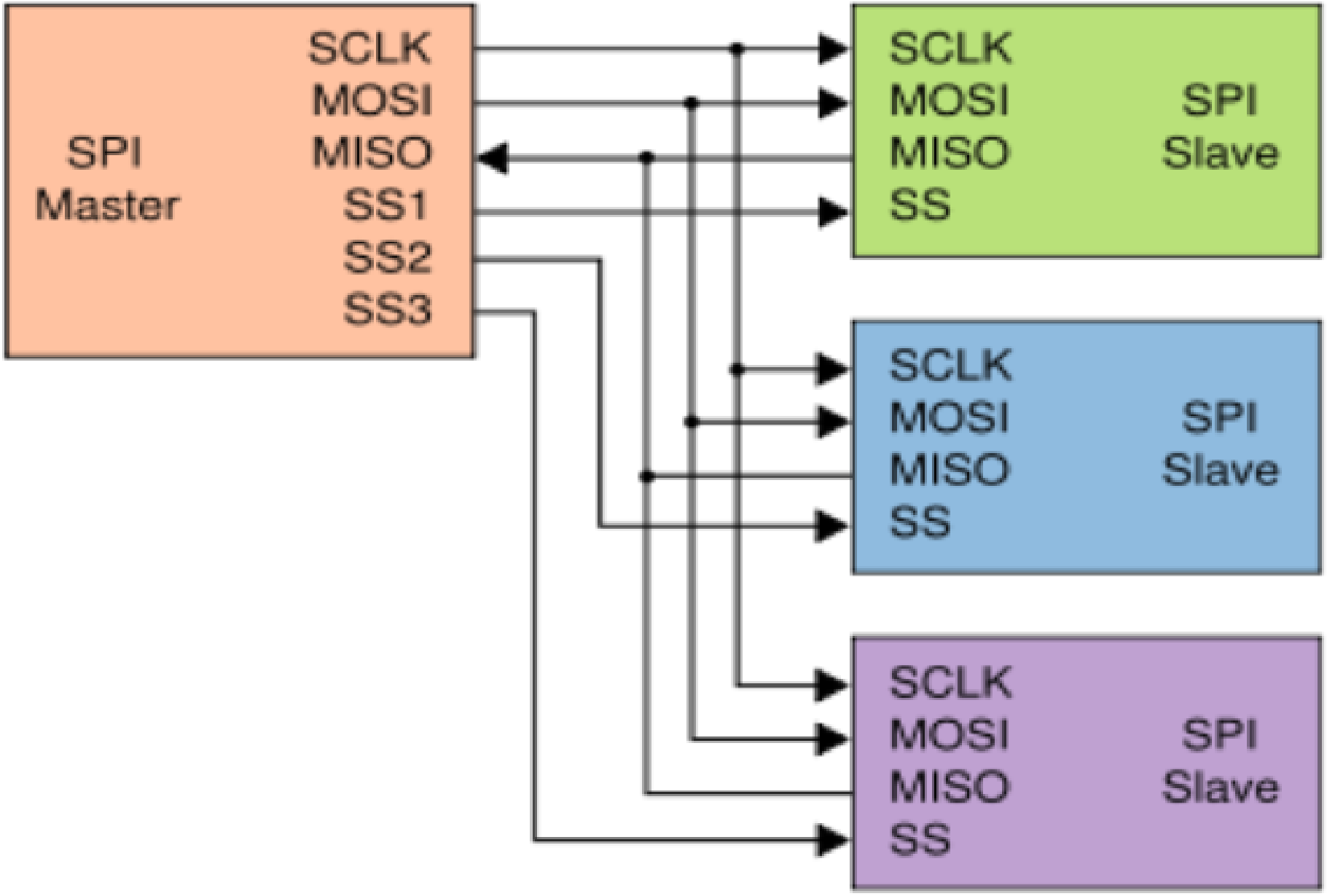
Data Validity & Transition

- At CPOL=0 the base value of the clock is zero
  - For CPHA=0, data is read on the clock's rising edge (low->high transition) and data is changed on a falling edge (high->low clock transition).
  - For CPHA=1, data is read on the clock's falling edge and data is changed on a rising edge.
- At CPOL=1 the base value of the clock is one (inversion of CPOL=0)
  - For CPHA=0, data is read on clock's falling edge and data is changed on a rising edge.
  - For CPHA=1, data is read on clock's rising edge and data is changed on a falling edge.

Modes

The combinations of polarity and phases are often referred to as modes which are commonly numbered according to the following convention:

Mode	CPOL	CPHA	Data Sampling	Data Transition
0	0	0	Rising Edge	Falling Edge
1	0	1	Falling Edge	Rising Edge
2	1	0	Falling Edge	Rising Edge
3	1	1	Rising Edge	Falling Edge

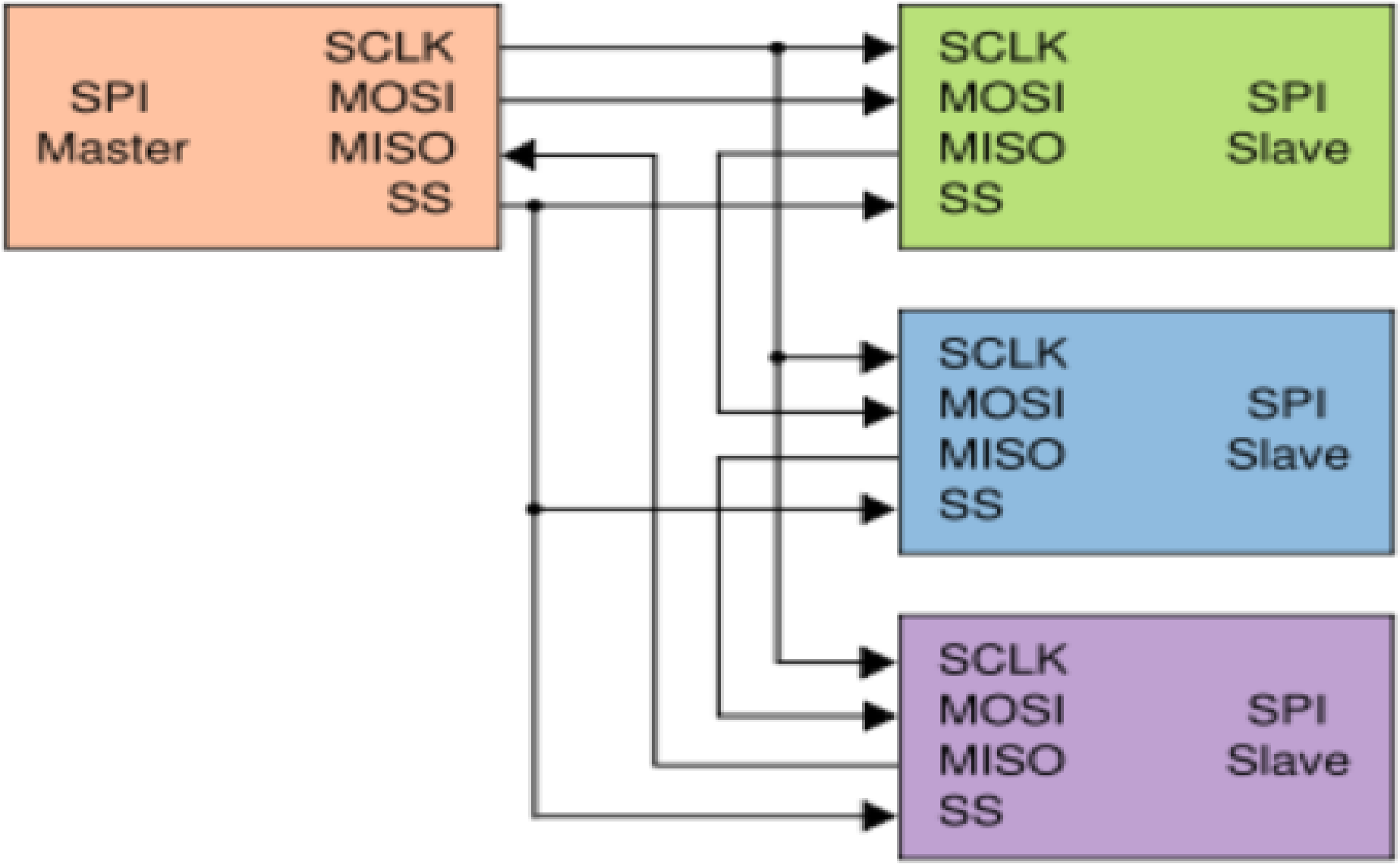


SPI bus: single master and three slaves

Example of Full-Duplex SPI Slaves

- MCP3204 – SPI Based ADC
- 25LC512 -- SPI BasedEEPROM
- MCP 2515 – SPI Based CAN Controller
- PCF8833 -- SPI Based GLCD controller for
- 48 x 48 LCD Screen
- MCP23S16/17 – SPI Based Port Expander
- ENC28J60 -- SPI Ethernet Controller
- Etc…

Daisy chain SPI configuration



SPI bus: one master and three daisy-chained slaves

Pros and cons of SPI

Advantages

- Full duplex communication for higher throughput
- Higher throughput than I^2C
- No addressing means reduced overhead which can be advantageous in small number (if not single) slaves

Disadvantages

- Requires more pins on IC packages:
  - Half duplex would require one fewer pin (possible in newer controllers with bidirectional mode; important slaves like EEPROMs tristate their outputs when receiving data from the master and don't care about the input when sending data back)
  - No addressing requires chip selects
- No hardware flow control
- No slave acknowledgment (the master could literally be "talking" to nothing and not know it)
- Is not a multi-master bus

I2C v/s SPI

I2C	SPI
Speed limit varies from 100kbps, 400kbps, 1mbps, 3.4mbps depending on i2c version.	More than 1mbps, 10mbps till 100mbps can be achieved.
Half duplex synchronous protocol	Full Duplex synchronous protocol
Support Multi master configuration	Multi master configuration is not possible
Acknowledgement at each transfer	No Acknowledgement
Require Two Pins only SDA, SCL	Require separate MISO, MOSI, CLK & CS signal for each slave.
Addition of new device on the bus is easy	Addition of new device on the bus is not much easy a I2C
More Overhead (due to acknowledgement, start, stop)	Less Overhead
Noise sensitivity is high	Less noise sensitivity